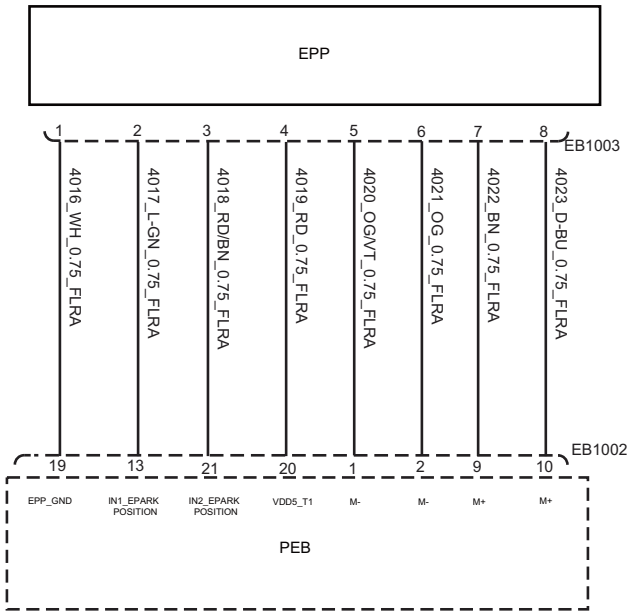
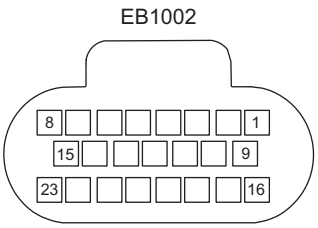
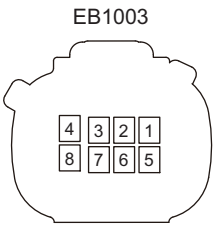
P1A81-71、P1A82-7C

Fault diagnosis code
P1A81-71：EPP motor stall
P1A82-7C：EPP system overtime fault
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • EPP fault • The EPP controller circuit in the PEU is faulty
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P1A81-71
EPP motor current greater than 6A or less than 15A and EPP operation, the sensor position is basically unchanged, reported locked rotor fault (only detected when the motor speed is less than 500rpm
P1A82-7C
If the EPP runs for more than 3s, it will not report a fault and will not retry
To set the effect of a fault code condition
Parking system malfunction

Circuit diagram





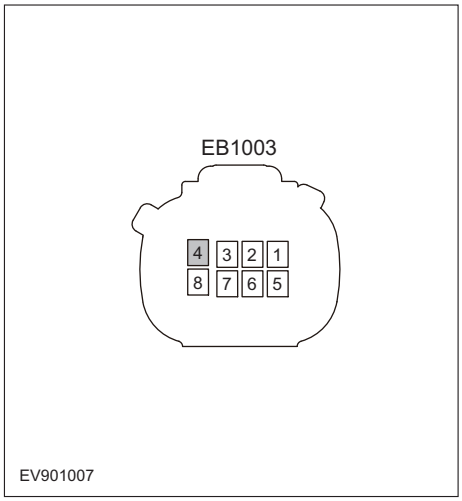


EV901006

Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of PEU for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the PEU with diagnostic apparatus. 4 If there is any fault code except for P1A81-71,P1A82-7C?
		YES → Refer to:DTC Summary list(PEU).
NO ↓		
Step 3	Perform initialization.	
		1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 The governor was initialized EPP with the diagnostic instrument. 5 EPP can the governor be initialized and is the governor restored?
		NO → Repair or change the EPP governor.
YES ↓		
Step 4	Perform EPP motor tests.	
		1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 Use the diagnostic instrument to perform EPP motor function test operation on the EPP controller. (If this function is present in the diagnostic) 5 Is the EPP motor functioning properly?
		NO → Repair or change the EPP module.

YES ↓

Step 5	Check the power supply circuit of EPP circuit.



1 Turn the ignition switch to OFF.

2 Disconnect the harness plug EB1003 of EPP.

3 Turn the ignition switch to ON.

4 Measure the voltage between the harness terminal of EPP and ground.

Measuring circuit: voltage between terminal 4 on plug EB1003 and ground.

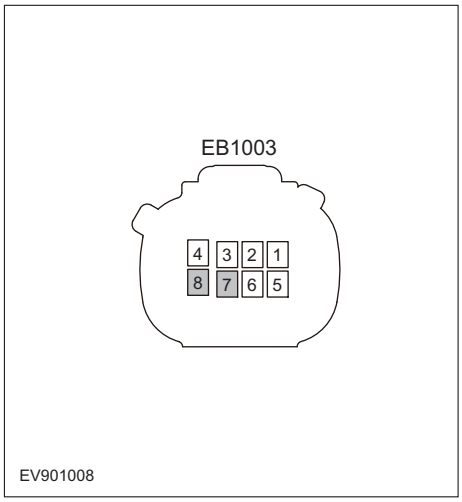
Standard value:4.8~5.2V

5 Check whether the result is normal or not?

NO → Repair or replace the faulted parts.

YES ↓

Step 6	Check the supply power of EPP when it is started or operating.



1 Turn the ignition switch to OFF.

2 Disconnect the harness plug EB1003 of EPP.

3 Turn the ignition switch to ON.

4 Turn on the EPP control switch to make the EPP motor in working condition.

5 Measure the voltage between the harness terminal of EPP and ground.

Measuring circuit: voltage between terminal 7 on plug EB1003 and ground.

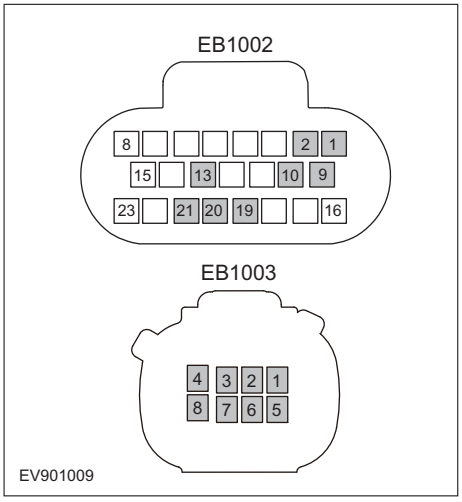
Measuring circuit: voltage between terminal 8 on plug EB1003 and ground.

Standard value:10~14V

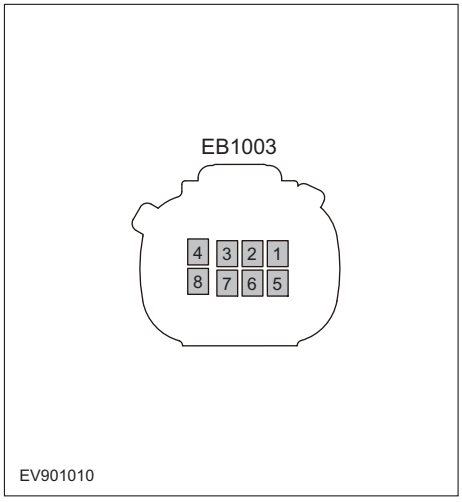
6 Check whether the result is normal or not?

NO → Repair or replace the faulted parts.

YES ↓

Step 7	Check the EPP circuit(open).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1002 of PEU module. 3 Measure the resistance between the harness terminal of EPP and the PEU. Measuring circuit: resistance between terminal 1 on plug EB1003 and terminal 19 on plug EB1002. Measuring circuit: resistance between terminal 2 on plug EB1003 and terminal 13 on plug EB1002. Measuring circuit: resistance between terminal 3 on plug EB1003 and terminal 21 on plug EB1002. Measuring circuit: resistance between terminal 4 on plug EB1003 and terminal 20 on plug EB1002. Measuring circuit: resistance between terminal 5 on plug EB1003 and terminal 1 on plug EB1002. Measuring circuit: resistance between terminal 6 on plug EB1003 and terminal 2 on plug EB1002. Measuring circuit: resistance between terminal 7 on plug EB1003 and terminal 9 on plug EB1002. Measuring circuit: resistance between terminal 8 on plug EB1003 and terminal 10 on plug EB1002. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 8	Check the circuit of EPP (short).



1 Turn the ignition switch to OFF.

2 Measure the resistance between the harness terminal of EPP and ground.

Measuring circuit: resistance between terminal 1 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 2 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 3 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 4 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 5 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 6 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 7 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 8 on plug EB1002 and ground.

Standard value: ∞

3 Measure the voltage between the harness terminal of EPP and ground.

Measuring circuit: voltage between terminal 1 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 2 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 3 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 4 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 5 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 6 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 7 on plug EB1002 and ground.

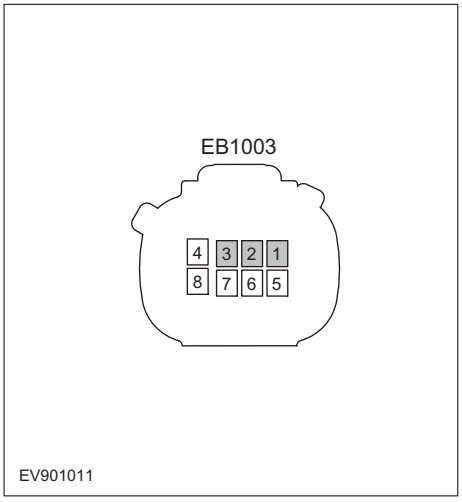
Measuring circuit: voltage between terminal 8 on plug EB1002 and ground.

Standard value: ≈ 0V

4 Check whether the result is normal or not?

NO → Repair or replace the faulted parts.

YES ↓

Step 9	Check the EPP (components).
	1 Measure the resistance of the EPP terminal directly. Measuring circuit: resistance between terminal 2 and terminal 1 on plug EB1003. Measuring circuit: the resistance between terminal 3 and 1 on plug EB1003. **Standard value:** 2 Check whether the result is normal or not?
	NO → Replace the EPP.
YES ↓	
Step 10	Check EPP controller circuit.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the PEU end cover. 4 Check EPP controller circuit. 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 11	Test whether PEU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1002 of PEU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. 4 Fly out of pins-install new pins as needed. 5 Reconnect the PEU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 6 Run the system and determine if the problem persists?
	YES → Change the PEU Module
NO ↓	

At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.
Next Step ↓
Diagnosis end.